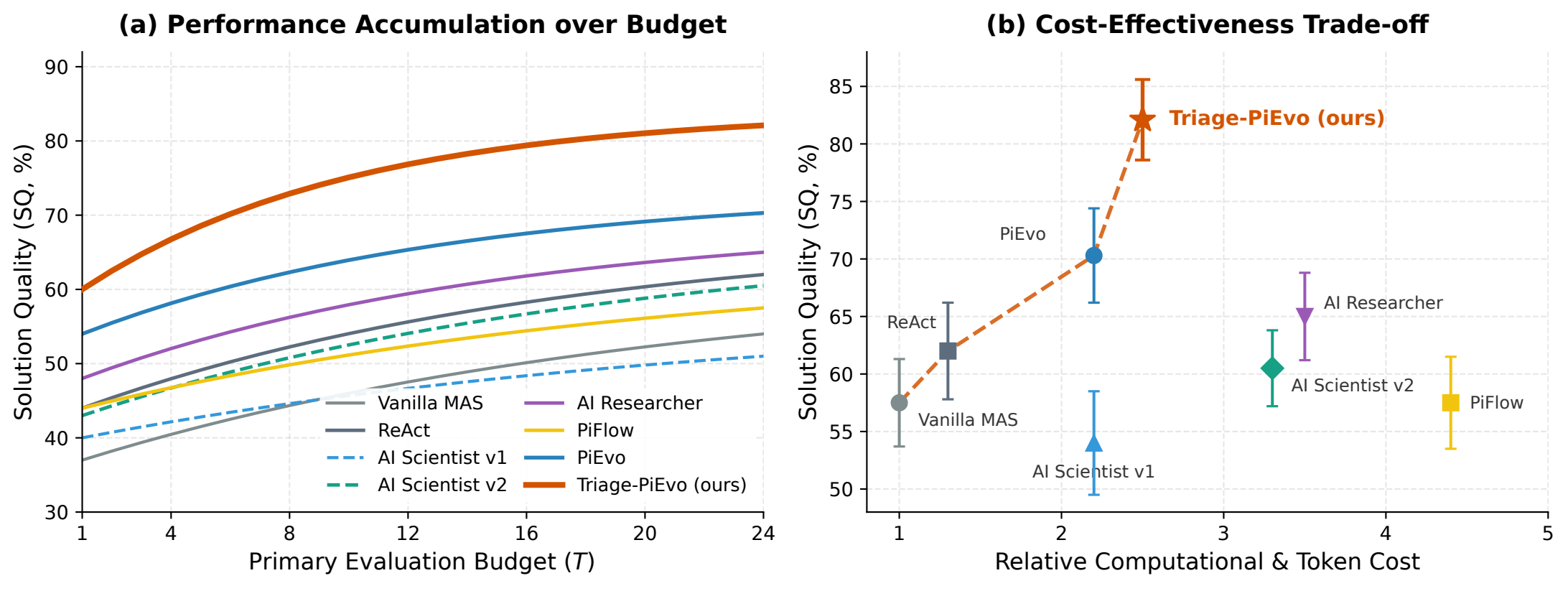
(a) Performance Accumulation over Budget
Solution Quality (SQ, %)
Primary Evaluation Budget (T)
Vanilla MAS
ReAct
AI Scientist v1
AI Scientist v2
AI Researcher
PiFlow
PiEvo
Triage-PiEvo (ours)
(b) Cost-Effectiveness Trade-off
Solution Quality (SQ, %)
Relative Computational & Token Cost
Triage-PiEvo (ours)
PiEvo
ReAct
Vanilla MAS
AI Scientist v1
AI Researcher
AI Scientist v2
PiFlow